

FileEditSelectionViewGoRun

AI Lab Programs

AI_Lab_Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

AI_Program_8

README.md

AI_Program_6 UAI_Program_7 UAI_Program_8 1, UAI_Program_4

AI_Program_8 > Perceptron > activate > x

```
1 import numpy as np
2
3 # --- PART A: Perceptron ---
4 class Perceptron:
5
6     def __init__(self, input_size, lr=0.1, epochs=10):
7         self.weights = np.zeros(input_size + 1) # Includes bias
8         self.lr = lr
9         self.epochs = epochs
10
11     def activate(self, x):
12         return 1 if x >= 0 else 0
13
14     def train(self, X, y):
15
16         for _ in range(self.epochs):
17
18             for xi, target in zip(X, y):
```

Problems 1OutputDebug ConsoleTerminalPorts

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_8"

--- Training Perceptron (AND Gate) ---

Prediction for [1, 1]: 1

--- Training Backpropagation (XOR Gate) ---

Final outputs after 5000 epochs:

[[0.31156095]

[0.65151416]

[0.64764441]

[0.42283689]]

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Ln 11, Col 24Spaces: 4UTF-8CRLF{} PythonGo Live

FileEditSelectionViewGoRun

AI Lab Programs

AI_Lab_Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

AI_Program_8

README.md

AI_Program_6 U AI_Program_7 U AI_Program_8 1, U AI_Program_4

AI_Program_8 > Perceptron > activate > x

4 class Perceptron:

14 def train(self, X, y):

19

20 inputs = np.insert(xi, 0, 1) # Insert bias unit

21

22 prediction = self.activate(

23 | np.dot(inputs, self.weights)

24 |)

25

26 self.weights += (

27 | self.lr * (target - prediction) * inputs

28 |)

29

30 def predict(self, X):

31

32 inputs = np.insert(X, 0, 1)

Problems 1 Output Debug Console Terminal Ports

Code

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_8"

--- Training Perceptron (AND Gate) ---

Prediction for [1, 1]: 1

--- Training Backpropagation (XOR Gate) ---

Final outputs after 5000 epochs:

[[0.31156095]

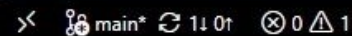
[0.65151416]

[0.64764441]

[0.42283689]]

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Ln 11, Col 24 Spaces: 4 UTF-8 CRLF Python Go Live



FileEditSelectionViewGoRun...AI Lab Programs

AI Lab Programs

.dist

.gitattributesAI_Program_1AI_Program_1(24BECS124).pdfAI_Program_2AI_Program_2(24BECS124).pdfAI_Program_3AI_Program_3(24BECS124).pdfAI_Program_4AI_Program_4(24BECS124).pdfAI_Program_5AI_Program_6AI_Program_7AI_Program_8README.md

AI_Program_6 UAI_Program_7 UAI_Program_8 1, UAI_Program_4

AI_Program_8 > train_backpropagation

45 def train_backpropagation(X, y, epochs=5000, lr=0.1):
50 wh = np.random.uniform(
51 size=(input_layer_size, hidden_layer_size)
52)
53
54 wout = np.random.uniform(
55 size=(hidden_layer_size, output_layer_size)
56)
57
58 for _ in range(epochs):
59
60 # Forward pass
61 hidden_layer_input = np.dot(X, wh)
62
63 hidden_layer_activations = sigmoid(
64 hidden_layer_input
65)
66
67 output_layer_input = np.dot(
 hidden_layer_activations,

Problems 1OutputDebug ConsoleTerminalPorts

Code + - - X

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_8"
--- Training Perceptron (AND Gate) ---
Prediction for [1, 1]: 1

--- Training Backpropagation (XOR Gate) ---
Final outputs after 5000 epochs:
[[0.31156095]
[0.65151416]
[0.64764441]
[0.42283689]]

Ln 65, Col 10Spaces: 4UTF-8CRLF{ }PythonGo Live

File Edit Selection View Go Run ...

AI Lab Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6 U

AI_Program_7 U

AI_Program_8 1, U

README.md

AI_Program_6 U

AI_Program_7 U

AI_Program_8 1, U

AI_Program_4

AI_Program_8 > train_backpropagation

45 def train_backpropagation(X, y, epochs=5000, lr=0.1):

71 | predicted_output = sigmoid(

72 | | output_layer_input

73 |)

74 |

75 | # Backpropagation

76 | error = y - predicted_output

77 |

78 | d_predicted_output = (

79 | | error * sigmoid_derivative(predicted_output)

80 |)

81 |

82 | hidden_layer_error = d_predicted_output.dot(

83 | | wout.T

84 |)

85 |

86 | d_hidden_layer = (

87 | | hidden_layer_error

88 | | * sigmoid_derivative(hidden_layer_activations)

Problems 1

Output

Debug Console

Terminal

Ports

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_8"

--- Training Perceptron (AND Gate) ---

Prediction for [1, 1]: 1

--- Training Backpropagation (XOR Gate) ---

Final outputs after 5000 epochs:

[[0.31156095]

[0.65151416]

[0.64764441]

[0.42283689]]

Ln 65, Col 10

Spaces: 4

UTF-8

CRLF

{ }

Python

Go Live

FileEditSelectionViewGoRun...

AI Lab Programs

AI Lab Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

AI_Program_8

README.md

AI_Program_6 UAI_Program_7 UAI_Program_8 1, UAI_Program_4

AI_Program_8 > train_backpropagation

45 def train_backpropagation(X, y, epochs=5000, lr=0.1):

101

102 return predicted_output

103 # --- Example Usage ---

104

105 if __name__ == "__main__":

106

107 print("--- Training Perceptron (AND Gate) ---")

108

109 X_and = np.array([

110 [0, 0],

111 [0, 1],

112 [1, 0],

113 [1, 1]

114])

115

116 y_and = np.array([0, 0, 0, 1])

117

118 p = Perceptron(input_size=2)

119

120 p.train(X_and, y_and)

Problems 1OutputDebug ConsoleTerminalPorts

Code + - - - - -

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_8"

--- Training Perceptron (AND Gate) ---

Prediction for [1, 1]: 1

--- Training Backpropagation (XOR Gate) ---

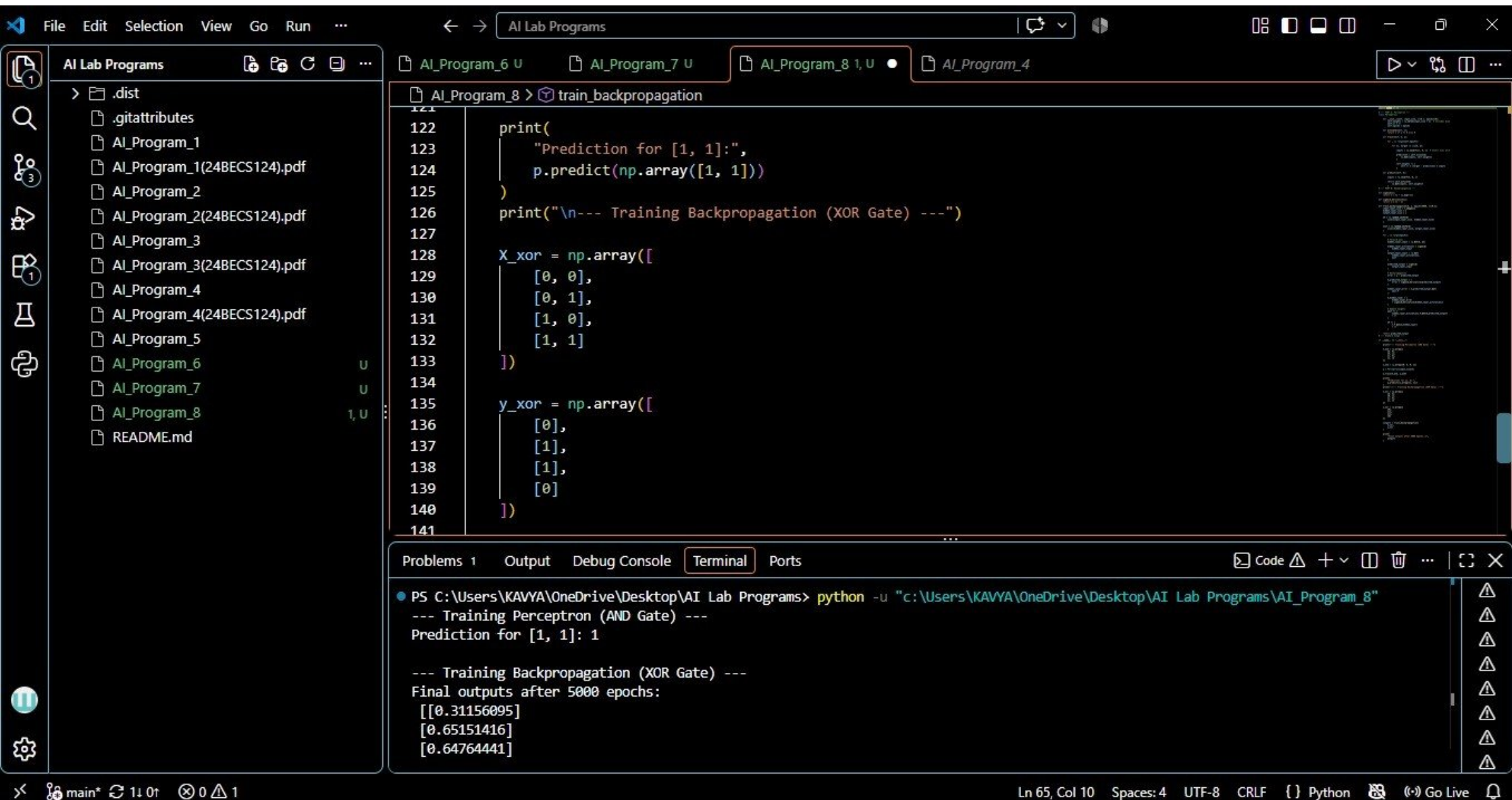
Final outputs after 5000 epochs:

[[0.31156095]

[0.65151416]

[0.64764441]

Ln 65, Col 10Spaces: 4UTF-8CRLF{ }PythonGo Live



File Edit Selection View Go Run

AI Lab Programs

AI Lab Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

AI_Program_8

README.md

AI_Program_6 U

AI_Program_7 U

AI_Program_8 1, U

AI_Program_4

AI_Program_8 > train_backpropagation

132 [1, 1]

133]]

134

135 y_xor = np.array([

136 [0],

137 [1],

138 [1],

139 [0]

140]]

141

142 outputs = train_backpropagation(

143 X_xor,

144 y_xor

145)

146

147 print(

148 "Final outputs after 5000 epochs:\n",

149 outputs

150)

Problems 1

Output

Debug Console

Terminal

Ports

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_8"

--- Training Perceptron (AND Gate) ---

Prediction for [1, 1]: 1

--- Training Backpropagation (XOR Gate) ---

Final outputs after 5000 epochs:

[[0.31156095]

[0.65151416]

[0.64764441]

Ln 65, Col 10 Spaces: 4 UTF-8 CRLF {} Python Go Live